



Title: Relationship between various customer attributes and their banking behaviour using non-parametric statistical methods

Mrunal Dhiwar, Priya Yadav ,Shem Basumatary, Surbhi Gari
(Supervisor: **Dr. Sharmishtha Mitra**)

0.1 ACKNOWLEDGEMENT

First and foremost, we thank our project supervisor, **Dr. Sharmishtha Mitra**, for her invaluable guidance, encouragement, and insightful feedback throughout the course of this project. Her expertise and mentorship have been instrumental in deepening our understanding of non-parametric statistical methods and their applications to Bank Churners data. We also extend our appreciation to our team members: **Mrunal Dhiwar, Priya Yadav, Shem Basumatary and Surbhi Gari**. Their hard work, dedication, and collaborative spirit were crucial in analyzing the data. *We would like to express our heartfelt gratitude to everyone who contributed to the success of this project.*

*First and foremost, we thank our project supervisor, **Dr. Sharmishtha Mitra**, for her invaluable guidance, encouragement, and insightful feedback throughout the course of this project. Her expertise and mentorship have been instrumental in deepening our understanding of non-parametric statistical methods and their applications to Bank Churners data. We also extend our appreciation to our team members: **Mrunal Dhiwar, Priya Yadav, Shem Basumatary and Surbhi Gari**. Their hard work, dedication, and collaborative spirit were crucial in analyzing the data.*

Contents

0.1	ACKNOWLEDGEMENT	i
0.2	INTRODUCTION	iii
0.3	PROBLEM STATEMENT	iv
0.4	DATASET DESCRIPTION	v
0.5	NON-PARAMETRIC TESTS PERFORMED	vii
0.5.1	Chi-Square Tests	vii
0.5.2	Mann-Whitney U Test	ix
0.5.3	Spearman's Rank Correlation	xi
0.5.4	Kolmogorov- Smirnov Test	xiii
0.5.5	Mood's Median Test	xvi

0.2 INTRODUCTION

Non-parametric tests are statistical tests that do not make assumptions about the underlying distribution of the data. Unlike parametric tests, which rely on assumptions like normality or equal variances, non-parametric tests are more flexible and can be used when the data do not meet the assumptions of parametric tests. These tests are particularly useful when the data are not normally distributed, the sample size is small, or when dealing with ordinal data or ranks.

Some common non-parametric tests include:

- *Sign test: Used to assess the difference between paired data when the data are ordinal or not normally distributed.*
- *Wilcoxon signed-rank test: A paired sample test that compares two related samples to determine if their population mean ranks differ.*
- *Mann-Whitney U test: A test used to determine whether there is a difference between two independent groups.*
- *Kruskal-Wallis test: An extension of the Mann-Whitney test for comparing more than two independent groups.*
- *Friedman test: A non-parametric alternative to the repeated measures ANOVA, used to test differences across multiple related groups.*

Non-parametric tests are widely used across various fields, such as biology, psychology, sociology, and economics. They offer a valuable tool for analyzing data when parametric assumptions are not met, especially in cases where traditional statistical methods are difficult or impractical to apply.

0.3 PROBLEM STATEMENT

In the competitive banking industry, customer retention is a critical factor for maintaining profitability and growth. Customer attrition, or "churn," occurs when customers close their accounts or discontinue their banking relationships. High attrition rates can significantly impact revenue, as acquiring new customers is often more costly than retaining existing ones. This study aims to identify key factors that influence customer attrition and to analyze the differences between customers who remain loyal and those who leave. By examining demographic information, credit behavior, transaction history, and relationship metrics, the objective is to determine which factors most strongly correlate with customer attrition and provide actionable insights for targeted retention strategies.

0.4 DATASET DESCRIPTION

The **BankChurners** dataset, consisting of **10,127 records and 23 attributes**, provides insights into customer characteristics and behaviors relevant to banking churn analysis. Below is a description of each attribute:

1. **CLIENTNUM**: Unique customer identifier.
2. **Attrition_Flag**: Categorical variable indicating customer retention status—whether the customer is an **Existing Customer** or an **Attrited Customer**.
3. **Customer_Age**: Integer representing the age of the customer.
4. **Gender**: Categorical variable representing the customer's gender.
5. **Dependent_count**: Number of dependents linked to the customer.
6. **Education_Level**: Customer's highest level of education, e.g., Graduate, High School.
7. **Marital_Status**: Customer's marital status, including options like Single, Married, etc.
8. **Income_Category**: Income bracket of the customer, such as **Less than \$40K**, **\$60K - \$80K**, etc.
9. **Card_Category**: Type of credit card held by the customer, e.g., Blue, Silver.
10. **Months_on_book**: Number of months the customer has maintained a relationship with the bank.
11. **Total_Relationship_Count**: Total count of banking products associated with the customer.
12. **Months_Inactive_12_mon**: Number of months the customer was inactive in the past year.
13. **Contacts_Count_12_mon**: Number of contacts (e.g., customer service calls) within the last 12 months.
14. **Credit_Limit**: Credit limit available on the customer's card.
15. **Total_Revolving_Bal**: Total outstanding balance that is revolving on the card.
16. **Avg_Open_To_Buy**: Average amount available for purchases on the customer's credit line.
17. **Total_Amt_Chng_Q4_Q1**: Proportionate change in the total transaction amount between Q4 and Q1.

18. **Total_Trans_Amt:** *Total amount of all transactions in the past 12 months.*

19. **Total_Trans_Ct:** *Total count of all transactions in the past 12 months.*

20. **Total_Ct_Chng_Q4_Q1:** *Proportionate change in transaction count between Q4 and Q1.*

21. **Avg_Utilization_Ratio:** *Average utilization ratio of the credit line, indicating the percentage of the credit limit used.* 22-23. **Naive_Bayes_Classifier_....:** *Probabilistic scores from a Naive Bayes classifier for attrition prediction.*

Data Characteristics

- **Categorical Features:** *Gender, Education_Level, Marital_Status, Income_Category, Card_Category.*
- **Numerical Features:** *Customer_Age, Dependent_count, Credit_Limit, Total_Relationship_Count, Months_Inactive_12_mon, Contacts_Count_12_mon, Total_Revolving_Bal, Avg_Open_To_Buy, Total_Trans_Amt, Total_Trans_Ct, Total_Amt_Chng_Q4_Q1, Total_Ct_Chng_Q4_Q1, Avg_Utilization_Ratio.*

Purpose

The dataset facilitates analysis of customer churn by examining demographic factors, credit usage, transaction behavior, and other relationship metrics with the bank. Through statistical and machine learning techniques, the data can help identify patterns and predictors of customer attrition, informing strategies for retention.

0.5 NON-PARAMETRIC TESTS PERFORMED

0.5.1 Chi-Square Tests

- The Chi-Square test of independence assesses whether there is a significant association between two categorical variables by comparing the frequency of each category of one variable across the categories of the other.
- The data is organized in a contingency table, with rows representing categories of one variable and columns representing categories of the other.

IMPLEMENTATION:

- We conducted chi-square tests to examine the association between the target variable, **Attrition_Flag**, and various customer attributes such as **Gender**, **Education Level**, **Card Category**, **Income Category**, and **Marital Status**.
- This analysis aimed to determine if there was a statistically significant relationship between customer attrition and these demographic variables.

Methodology

For each pair of variables, a contingency table was created to observe the distribution of categories across the two variables. Using the chi-square test of independence, we computed:

- **Chi-square Statistic:** Measures how much the observed frequencies in the contingency table deviate from the expected frequencies if the two variables were independent.
- **Degrees of Freedom (dof):** The number of values in the calculation of a statistic that are free to vary.
- **P-value:** Indicates the probability of observing a chi-square statistic as extreme as the one calculated if the null hypothesis of independence is true.

Test Statistic:

- The chi-square test statistic is calculated by comparing the observed frequencies in each cell of the contingency table with the expected frequencies (which would be observed if the variables were independent).
- The formula for the chi-square statistic (χ^2) is

$$\chi^2 = \sum \frac{(O - E)^2}{E} \quad (1)$$

Hypothesis:

- H_0 : The variables are independent, there is no relationship between the two categorical variables.
- H_A : The variables are dependent, there is a relationship between the two categorical variables.
- **Significant Level** (α : 0.05)

Decision Rule:

- We reject H_0 if the p -value was less than 0.05.

Result:

Dataset	Chi-Square Statistic	P-Value	Decision
Gender	13.8656	0.0001963	Reject
Education_Level	12.5112	0.0514891	Accept
Card_Category	2.2342	0.5252382	Accept
Income_Category	12.8322	0.0250024	Reject
Martial_Status	6.0560	0.1089126	Accept

Table 1: Chi-square tests result

Conclusion:

1. The target variable, "**Attrition_Flag**", is associated with the categorical features "**Income_Category**" and "**Gender**", indicating a significant relationship with customer churn.
2. This suggests that income levels and gender may play a role in influencing whether a customer decides to stay or leave, potentially making these features key predictors of attrition.
3. Identifying and analyzing these associations can provide valuable insights into the demographic factors that impact customer retention.

0.5.2 Mann-Whitney U Test

- *The Mann-Whitney U test (also known as the Wilcoxon rank-sum test) is a non-parametric test used to determine whether there is a difference in the central tendency (typically median) between two independent samples.*
- *It is particularly useful when the data do not necessarily follow a normal distribution.*

Assumptions

The test assumes that:

- *The two samples are independent.*
- *The data in each sample are drawn from a continuous distribution, so the possibility of ties $X_i = Y_j$ is generally ignored (or handled in a specific way if ties are present).*

*X : credit limit for **Existing Customers***

*Y : credit limit for **Attrited Customers***

In our case, we have tested for similar distribution of Attrition Flag v/s various features concerning:

1. *Credit Limit*
2. *Customer Age*
3. *Dependent count*
4. *Months on book*
5. *Total Relationship Count*
6. *Months Inactive 12 months*
7. *Contact Count 12 months*
8. *Total Revolving Balance*
9. *Average Open To Buy*
10. *Total Amount Change Quarter 4 to Quarter 1*
11. *Total Transaction Amount*
12. *Total Transaction Count*
13. *Total Count Change Q4 to Q1*
14. *Average Utilization Ratio*

Dataset	p- value	Decision
Credit_Limit	3.00784e-07	Reject
Customer_Age	0.0781	Fail to reject
Dependent_count	0.03473	Reject
Months_on_book	0.12366	Fail to reject
Total_Relationship_Count	2.90571e-51	Reject
Months_Inactive_12_mon	5.42311e-67	Reject
Contact_Count_12_mon	1.11088e-80	Reject
Total_Revolving_Bal	1.91567e-129	Reject
Avg_Open_To_Buy	0.00565	Reject
Total_Amt_Chng_Q4_Q1	1.06447e-24	Reject
Total_Trans_Amt	2.71900e-112	Reject
Total_Trans_Ct	0.0	Reject
Total_Ct_Chng_Q4_Q1	1.90824e-216	Reject
Avg_Utilization_Ratio	2.86909e-129	Reject

Test Statistic:

1. **Rank the Data:** Combine both samples X and Y into a single dataset, rank all values from smallest to largest, and assign average ranks in case of ties.
2. **Sum of Ranks:** Compute the sum of ranks for each group. Let R_X be the sum of ranks for sample X and R_Y be the sum of ranks for sample Y .
3. **Calculate U Statistics:** The Mann-Whitney U test calculates two possible U statistics, one for each group:

$$U_X = R_X - \frac{m(m+1)}{2}$$

$$U_Y = R_Y - \frac{n(n+1)}{2} \text{ where:}$$

- m is the sample size of X .
- n is the sample size of Y .

The Mann-Whitney U statistic is the smaller of U_X and U_Y : $U = \min (U_X, U_Y)$

Hypothesis:

- H_0 : The distributions of the two groups are identical.
- H_A : The distributions of the two groups are different.
- **Significant Level** (: 0.05)

Decision Rule: $p \text{ value} < 0.05$ then reject the null hypothesis.

Conclusion:

Significant Differences:

Most financial and behavioral metrics, like *Credit_Limit*, *Dependent_count*, *Total_Relationship_Count*, *Months_Inactive_12_mon*, and various transaction-related features, show significant differences between the two groups (*Attrited* vs. *Existing Customers*). These significant differences suggest that these features may have a meaningful impact on customer attrition and could be key predictors in a model.

Non-Significant Differences:

Customer_Age and *Months_on_book* show no significant difference between the two groups. This implies that age and tenure with the bank might not directly relate to attrition behavior in this dataset.

0.5.3 Spearman's Rank Correlation

- A random sample of n pairs

$$(X_1, Y_1), (X_2, Y_2), \dots, (X_n, Y_n)$$

is drawn from a bivariate population with Pearson product moment correlation coefficient r . In classical statistics, the estimate commonly used for r is the sample correlation coefficient defined as

$$R = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{[\sum_{i=1}^n (X_i - \bar{X})^2 \sum_{i=1}^n (Y_i - \bar{Y})^2]^{1/2}}$$

- Spearman's rank correlation measures the strength and direction of association between two ranked variables. How well the relationship between two variables could be represented using a monotonic function.

Hypothesis:

- H_0 : : There is no monotonic relationship between the two variables.
- H_A : A monotonic relationship exists between the two variables (either positive or negative).
- **Significant Level** (α : 0.05)

Interpretation:

- Positive values close to +1 indicate a strong positive monotonic relationship.
- Negative values close to -1 indicate a strong negative monotonic relationship.
- Values around zero indicate no correlation.

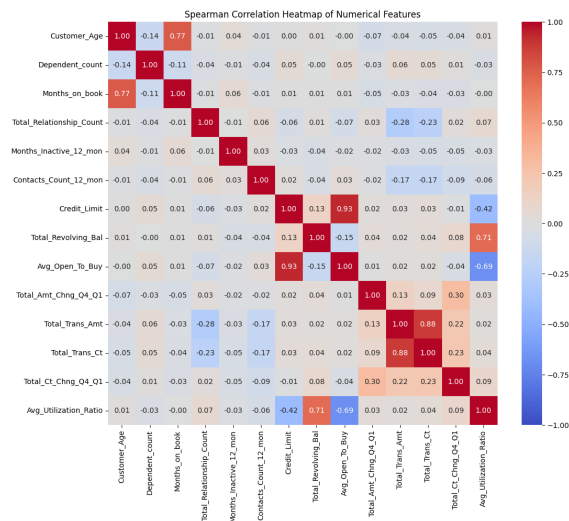


Figure 1: Spearman Correlation Heatmap

Conclusion: To test the relationship between the various feature:

- **Credit_Limit and Avg_Open_To_Buy (0.93):** Strong positive correlation indicates that customers with higher credit limits generally have more available credit.
- **Avg_Utilization_Ratio and Total_Revolving_Bal (0.71):** Higher revolving balances are associated with higher utilization ratios, suggesting more intensive credit usage.
- **Credit_Limit and Avg_Utilization_Ratio (-0.42):** Moderate negative correlation; customers with higher credit limits tend to have lower utilization ratios, potentially indicating lower credit dependency.
- **Total_Trans_Amt and Total_Trans_Ct (0.88):** High correlation between transaction amount and count, showing that customers who transact more frequently also spend more. This suggests high engagement among active users.
- **Total_Amt_Chng_Q4_Q1 and Total_Ct_Chng_Q4_Q1 (0.30):** Moderate positive correlation indicates a relationship between changes in transaction amount and transaction count from Q4 to Q1, reflecting spending pattern shifts over time.
- **Customer_Age and Months_on_book (0.77):** Older customers have been with the bank longer, suggesting age is positively associated with account longevity.
- **Dependent_count, Months_Inactive_12_mon, and Contacts_Count_12_mon:** These variables show weak correlations with other variables, indicating minimal association with other customer behaviors or credit metrics.

0.5.4 Kolmogorov- Smirnov Test

The Kolmogorov-Smirnov (KS) test is a nonparametric method for comparing the distribution functions of two samples or for evaluating how well a model fits observed data.

- **Comparing two samples:** *Determines if two samples come from the same distribution.*
- **Comparing a model and data:** *Assesses how well a model agrees with data by comparing the model's cumulative distribution function to the data points.*
- **Goodness-of-fit:** *Tests if a sample comes from a known distribution.*
- **Normality:** *Determines if a sample is normally shaped.*

The KS test uses a D-statistic to measure the maximum distance between the cumulative distribution functions of the two samples. It doesn't require the distribution of the data to be known in advance.

Methodology:

Formula: $D_{m,n} = \max_x |S_m(x) - S_n(x)|$ where,

- $S_m(x)$ = *Observed Cumulative frequency distribution of a random sample of n observations*
- $S_n(x)$ = *The theoretical frequency distribution*

Hypothesis:

- H_0 : *The two samples are drawn from the same distribution*
- H_A : *The two samples are drawn from different distributions.*

Decision Rule:

- *If the p-value is less than 0.05, we reject the null hypothesis, indicating a significant difference between churners and non-churners for that particular feature.*

Result:

Dataset	statistic	p-value	Significant
Customer _{Age}	0.035957	5.693640e-02	False
Credit _{Limit}	0.116053	1.769992e-16	True
Total _{RelationshipCount}	0.192427	1.012547e-44	True
Months _{on_book}	0.032094	1.170811e-01	False
Total _{RevolvingBal}	0.443710	8.093862e-244	True
Avg _{OpenToBuy}	0.132513	2.296391e-21	True
Total _{AmtChngQ4Q1}	0.167343	7.132385e-34	True
Total _{TransAmt}	0.484606	2.831369e-293	True
Total _{TransCnt}	0.550507	5.518713e-321	True
Total _{CtChngQ4Q1}	0.411329	2.581716e-208	True
Avg _{UtilizationRatio}	0.383815	1.398880e-180	True

Table 2: Chi-square tests result

Interpretation of Results:

- **Customer_Age:**

- No significant difference between churners and non-churners.
- **Conclusion:** Age alone does not distinguish churn behavior.

- **Credit_Limit:**

- Significant difference observed.
- **Conclusion:** Credit limits vary notably between churners and non-churners, which may influence churn.

- **Total_Relationship_Count:**

- Significant difference observed.
- **Conclusion:** A higher relationship count may correlate with customer retention, making it an important factor for churn prediction.

- **Months_on_book:**

- No significant difference observed.
- **Conclusion:** The length of customer tenure does not appear to influence churn.

- **Total_Revolving_Bal:**

- Significant difference observed.
- **Conclusion:** Differences in revolving balance patterns likely impact churn behavior.

- **Avg_Open_To_Buy:**

- *Significant difference observed.*
- **Conclusion:** *Available credit differs between churners and non-churners, influencing churn risk.*

- **Total_Amt_Chng_Q4_Q1:**

- *Significant difference observed.*
- **Conclusion:** *Changes in transaction amounts are notably different between groups, potentially reflecting churn patterns.*

- **Total_Trans_Amt:**

- *Significant difference observed.*
- **Conclusion:** *Total transaction amounts are a significant factor in churn prediction, possibly reflecting engagement levels.*

- **Total_Trans_Ct:**

- *Significant difference observed.*
- **Conclusion:** *The frequency of transactions varies notably, which may be indicative of retention behavior.*

- **Total_Ct_Chng_Q4_Q1:**

- *Significant difference observed.*
- **Conclusion:** *Quarterly changes in transaction count appear to relate to churn behavior.*

- **Avg_Utilization_Ratio:**

- *Significant difference observed.*
- **Conclusion:** *Credit utilization patterns vary significantly, with higher utilization potentially indicating higher churn risk.*

0.5.5 Mood's Median Test

- *Mood's Median Test is a non-parametric test used to assess whether the medians of two or more independent groups differ significantly.*
- *It's ideal for skewed data or data with outliers, as it doesn't require assumptions of normality or equal variances.*
- *By examining value distributions relative to the overall median, it evaluates if median differences could occur by chance, making it suitable for ordinal or non-normally distributed continuous data.*

Test Statistic for Mood's Median Test

- *Test statistic for this type of test is the Chi Squared statistic where we look for the Observed and expected frequencies.*

$$\chi^2 = \frac{\sum_{i=1}^k (O_{iAM} - E_{iAM})^2}{E_{iAM}} + \frac{\sum_{i=1}^k (O_{iBM} - E_{iBM})^2}{E_{iBM}}$$

- $(O_{i,A,M} = \text{Observed Frequencies of } i^{\text{th}} \text{ sample above Median})$
- $(O_{i,B,M} = \text{bserved Frequencies of } i^{\text{th}} \text{ sample above Median})$
- $(E_{i,A,M} = \text{Expected Frequencies of } i^{\text{th}} \text{ sample above Median})$
- $(E_{i,A,M} = \text{Expected Frequencies of } i^{\text{th}} \text{ sample above Median})$

Hypothesis:

- H_0 : *The medians of all groups are the same*
- H_A : *At least one group has a different median.*
- **Significant Level** ($\alpha : 0.05$)

Decision Rule:

If p-value less than 0.05 indicates that the difference in medians between churners and non-churners for that variable is statistically significant.

Result:

Dataset	Statistics	p-Value	Overall Median	Significant
Customer_Age	0.8525	0.3558	46.000	False
Credit_Limit	9.3830	0.0022	4549.000	True
Total_Relationship_Count	101.7827	6.20e-24	4.000	true
Months_on_book	0.2532	0.6148	36.000	False
Total_Revolving_Bal	473.1653	6.56e-105	1276.000	True
Avg_Open_To_Buy	0.0034	0.9534	3474.000	False
Total_Amt_Chng_Q4_Q1	45.7084	1.37e-11	0.736	True
Total_Trans_Amt	734.9715	7.43e-162	3899.000	True
Total_Trans_Ct	1270.0701	3.61e-278	67.000	True
Total_Ct_Chng_Q4_Q1	525.7681	2.35e-116	0.702	True
Avg_Utilization_Ratio	358.7246	5.34e-80	0.176	True

Table 3: Mood's median tests result

Conclusion:

- **Customer_Age:** No significant difference in median age between churners and non-churners (median = 46), indicating age is not a major factor in churn.
- **Credit_Limit:** Significant difference, with churners and non-churners showing distinct credit limits (median = 4549), suggesting varied access to credit.
- **Total_Relationship_Count:** Significant difference in the number of relationships with the bank (median = 4), with one group having more bank connections.
- **Months_on_book:** No significant difference in tenure (median = 36 months), indicating that tenure length does not strongly influence churn likelihood.
- **Total_Revolving_Bal:** Significant difference in revolving balance levels (median = 1276), hinting at differing credit behaviors between groups.
- **Avg_Open_To_Buy:** No significant difference in available credit (median = 3474), showing similar open-to-buy credit across groups.
- **Total_Amt_Chng_Q4_Q1:** Significant difference in transaction amount changes between quarters (median = 0.736), suggesting differing spending behavior changes.
- **Total_Trans_Amt:** Significant difference in total transaction amounts (median = 3899), reflecting varied spending levels between groups.
- **Total_Trans_Ct:** Significant difference in transaction counts (median = 67), indicating different transaction frequencies between churners and non-churners.
- **Total_Ct_Chng_Q4_Q1:** Significant difference in transaction count change (median = 0.702), pointing to distinct transaction frequency trends.

- **Avg_Utilization_Ratio:** *Significant difference in utilization ratio (median = 0.176), suggesting different credit usage patterns between groups.*